Remote Access and File Transfer Tools: Installation and Practical Use

Tools covered: FileZilla, WinSCP, PuTTY, MobaXterm

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Course: SEC 204: Threat Modelling and Security Monitoring Sessional

Submitted to: Md. Masud Rana

Date: 11 November 2026

1. Introduction

In system administration and cyber security work, we very often need to log in to a remote machine and move files to and from it. The Secure Shell (SSH) protocol makes this possible over an encrypted channel, and SFTP (SSH File Transfer Protocol) runs on top of SSH for secure file transfer. In this lab we installed four popular tools on a Windows host (FileZilla, WinSCP, PuTTY and MobaXterm) and used them to connect to a Kali Linux virtual machine. This report describes the lab setup, the installation process, how each tool was configured, and what we observed.

2. Objectives

- To install FileZilla, WinSCP, PuTTY and MobaXterm on Windows.
- To prepare a Kali Linux virtual machine as an SSH/SFTP server.
- To connect to the server from each tool using the SSH protocol on port 22.
- To compare the tools by purpose, interface and typical use cases.

3. Lab Environment

Item	Details
Client (host) machine	Windows PC, user profile *Lenovo*
Server (guest) machine	Kali Linux 7.1.5 (kali-amd64) running in Oracle VirtualBox
Network	Host-only adapter (eth1) with IP 192.168.56.102/24; NAT adapter (eth0) for internet
Server username	iftiur
Protocol and port	SSH / SFTP, TCP port 22
Host-side address	192.168.56.1 (seen in the PuTTY login message)

The `ip a` command inside Kali showed that interface *eth1* had the address 192.168.56.102, and this address was used as the host name in every tool below.

The VirtualBox Manager lists three virtual machines: Kali Linux (running), Wazuh v4.14.9 OVA (powered off) and Ubuntu Server (running). The Kali Linux VM is the SSH server used for the connections in this report.

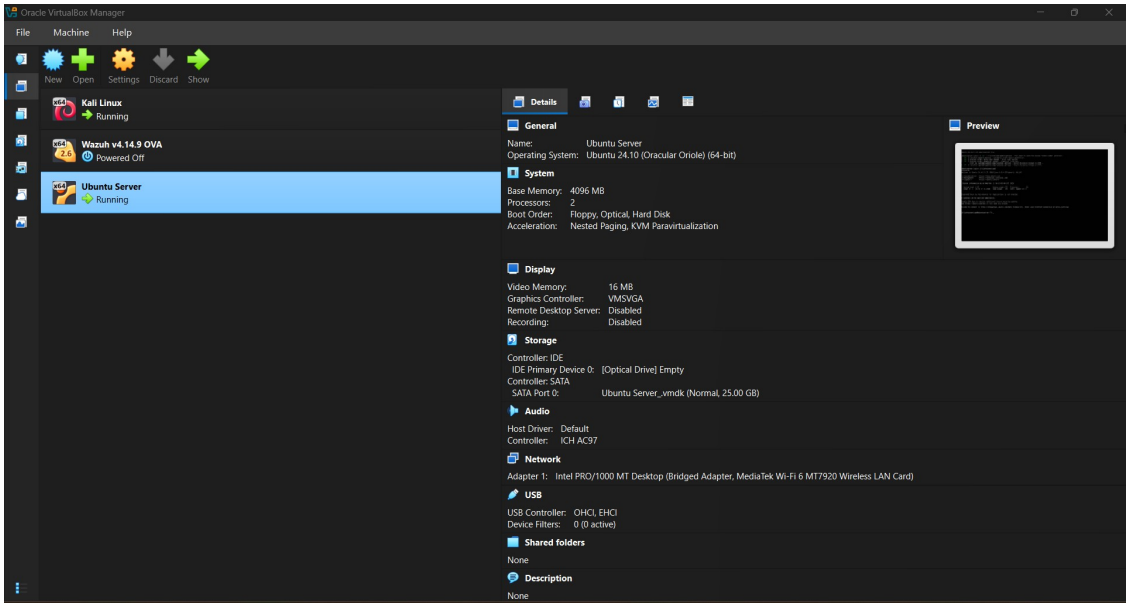


Figure 1: Oracle VirtualBox Manager showing the lab virtual machines and the Ubuntu Server settings.

The Ubuntu Server VM has 4096 MB memory, 2 processors, a 25 GB virtual disk and one network adapter in bridged mode (Wi-Fi). Its console shows Ubuntu 24.04.5 LTS with a successful login, so it can also be used as a second Linux server with the same tools.

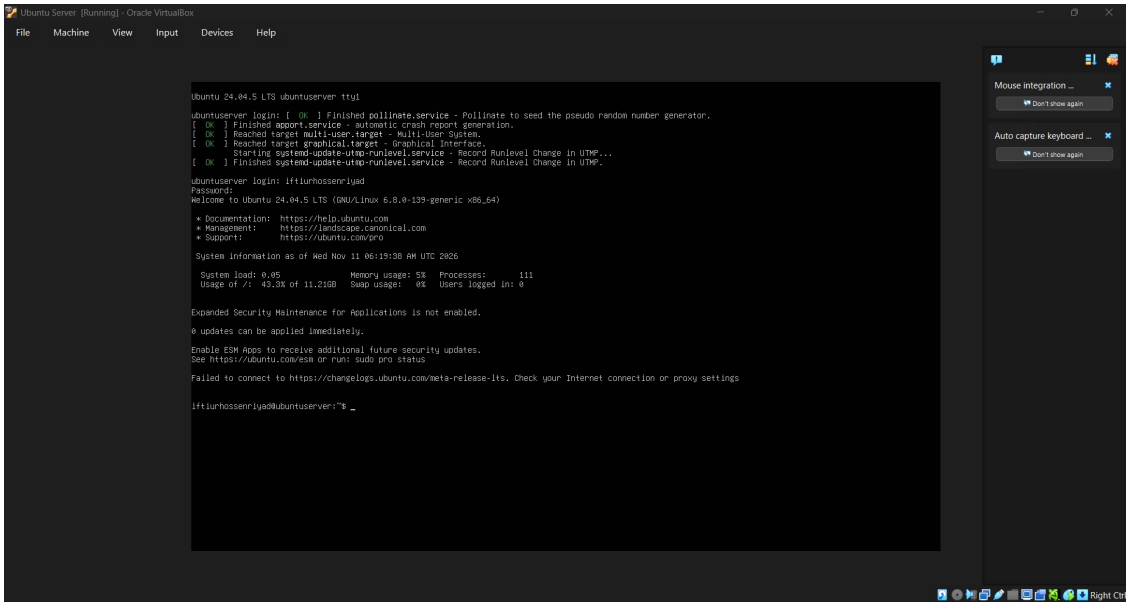


Figure 2: Ubuntu Server 24.04.5 LTS console after a successful login.

4. Preparing the Kali Linux Server

Before any client can connect, the SSH service must be installed and running on Kali. The standard commands are:

```
sudo apt update
sudo apt install openssh-server
sudo systemctl enable --now ssh
sudo systemctl status ssh
ip a
```

The last command is used to find the VM's host-only IP address. Because the VM uses a host-only network, the Windows host and the VM can reach each other directly, and the server is not exposed to the outside network.

5. Installation of the Tools

All four tools were installed on Windows using their official installers. The general procedure is the same for each:

1. Download the latest installer from the tool's official website.
2. Run the installer and accept the license agreement.
3. Choose the installation folder (the default is fine) and the shortcuts.
4. Finish the installation and launch the application.

Notes specific to each tool:

- **FileZilla:** Download the FileZilla Client (not the Server). During setup, decline any bundled optional software offered by the installer.
- **WinSCP:** Choose the *Typical installation* and the *Commander* interface (two panels) when asked.
- **PuTTY:** Install the 64-bit MSI package. It provides PuTTY, PuTTYgen and Pageant.
- **MobaXterm:** The Home (Personal) edition is free; the installer edition or the portable edition can be used. Our screen showed MobaXterm Personal Edition v26.6.

6. Step-by-Step Connection Procedure

Every tool uses the same server details: host `192.168.56.102`, port `22`, username `iftiur`. To find the host address, run `ip a` in the Kali terminal and read the `inet` value of *eth1* (visible on the left side of Figures 7 and 8).

6.1 FileZilla (SFTP)

1. Open FileZilla on Windows. The right panel shows *Not connected to any server*, and the left panel shows the local folder `C:\Users\Lenovo\`.
2. In the Quickconnect bar, type the host `192.168.56.102`, username `iftiur`, the password and port `22`.

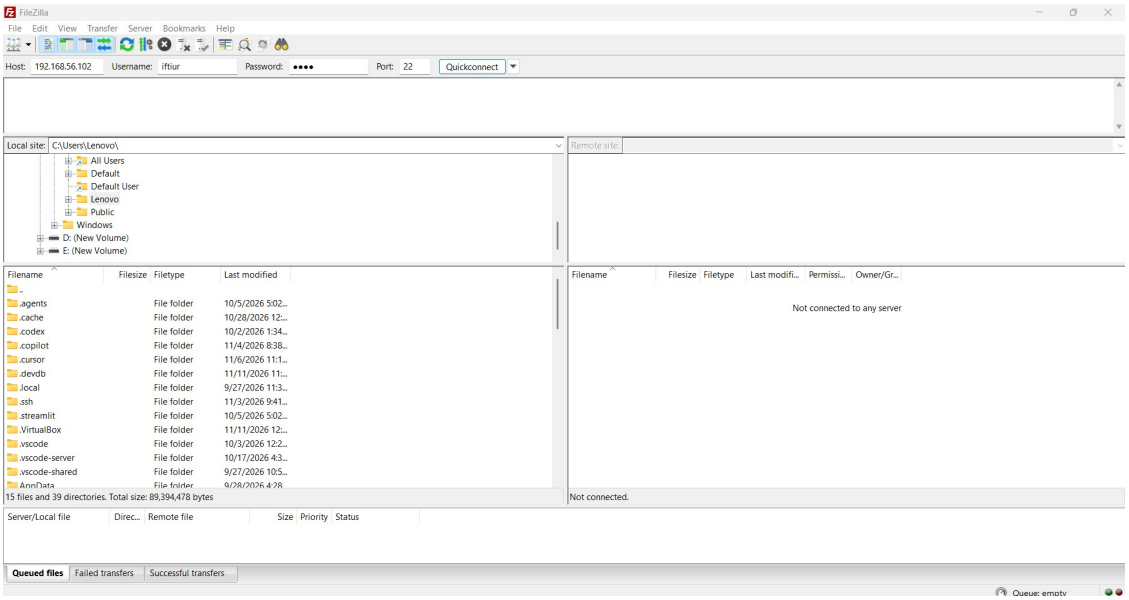


Figure 3: FileZilla with the server details entered, before connecting.

3. Click *Quickconnect*. When the host key prompt appears, check the fingerprint and accept it.
4. The status log confirms the connection: *Connecting to 192.168.56.102:22*, *Hostkey is trusted*, and *Directory listing of "/home/iftiur" successful*.
5. The right panel now shows the remote Kali home folder (`.config`, `.ssh`, `Desktop`, `Documents`, `Downloads`, `eduquiz`, `lab2` and others) with permissions and owner `iftiur`.

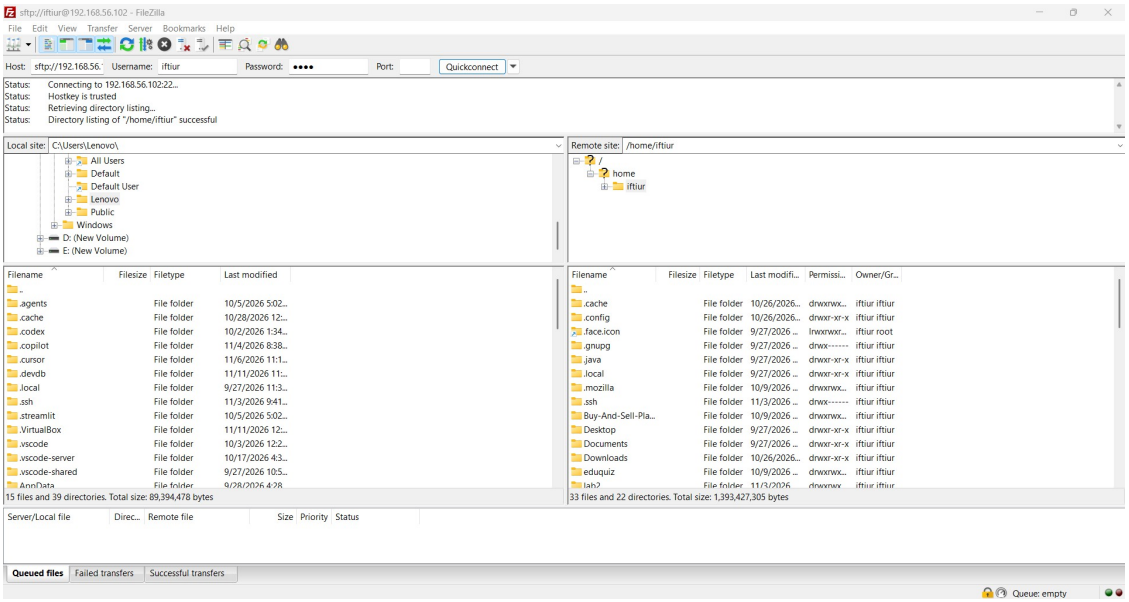


Figure 4: FileZilla connected; local files on the left, remote `/home/iftiur` on the right.

6. To transfer a file, drag it from one panel to the other. The queue at the bottom shows its progress.

6.2 WinSCP (SFTP)

1. Open WinSCP. The *Login* window appears.
2. Set *File protocol* to `SFTP`, *Hostname* to `192.168.56.102`, *Port number* to `22`, *Username* to `iftiur`, and enter the password.

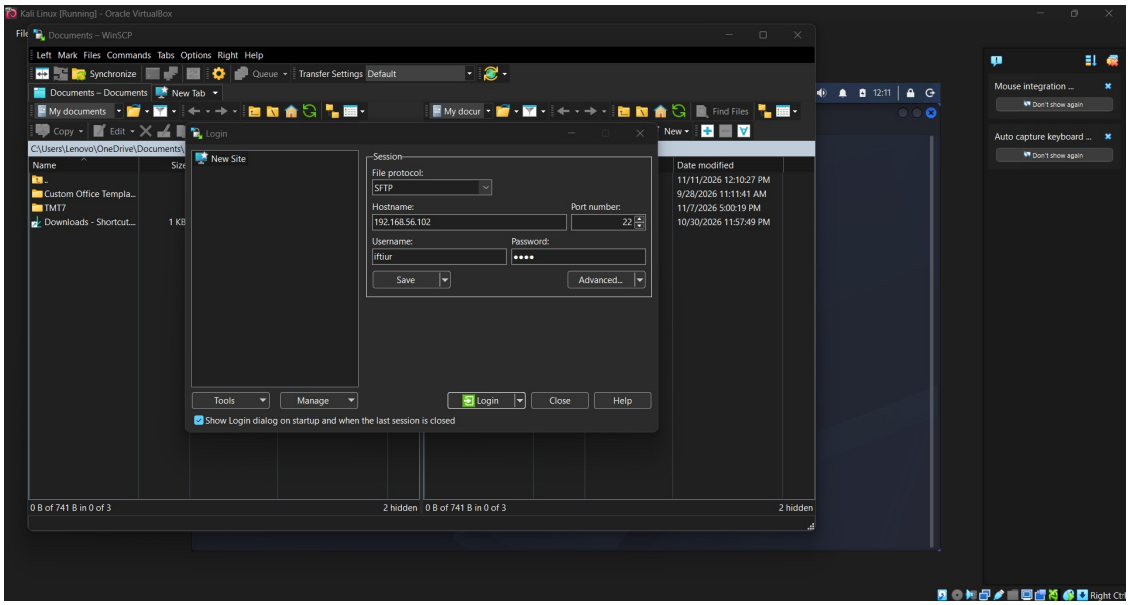


Figure 5: WinSCP login window filled with the Kali server details.

3. Click *Login* and accept the host key warning on the first connection.
4. The left panel shows the Windows folder `C:\Users\Lenovo\OneDrive\Documents\`, and the right panel shows `/home/iftiur/` with folders such as `Buy-And-Sell-Platform`, `Projects` and `Templates`, and files such as `auth.log`, `hello.sh`, `test.sh` and `vt_automation.py`.
5. The *Rights* and *Owner* columns show Linux permissions (for example `rwxr-xr-x`). The status bar shows `SFTP-3` and a lock icon, which means the session is encrypted.

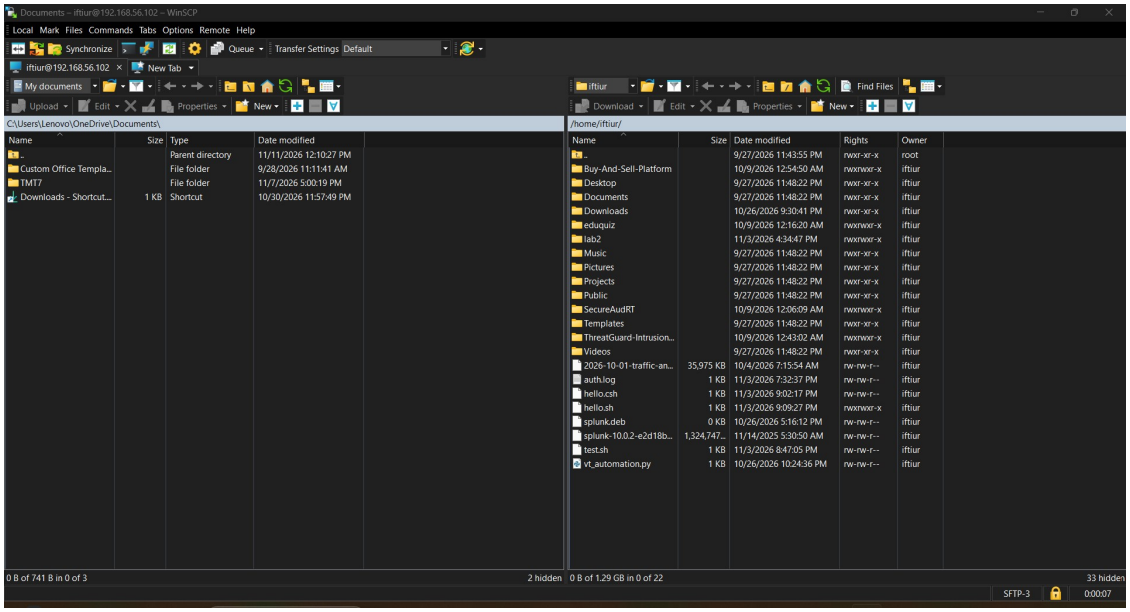


Figure 6: WinSCP connected to `/home/iftiur/` on the Kali VM.

6.3 PuTTY (SSH)

1. Open PuTTY. In the *Session* category, type `192.168.56.102` as the host name and `22` as the port.
2. Make sure the connection type is *SSH*, then click *Open*.

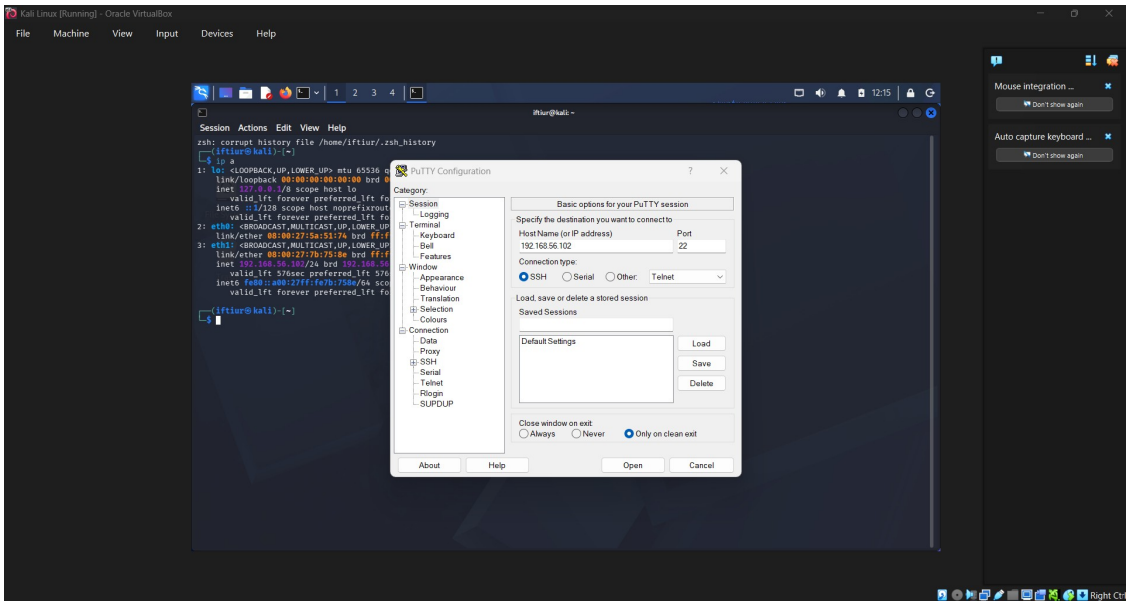


Figure 7: PuTTY Configuration window; the Kali terminal with the `ip a` output is behind it.

3. At the `login as:` prompt, type `iftiur`, then enter the password (it is not displayed while typing).
4. The Kali banner appears (Linux kali 7.1.5+kali-amd64) with the last login line *Tue Nov 10 15:45:10 2026 from 192.168.56.1*, which shows that the login came from the Windows host.

5. The prompt `(iftiur@kali)-[~]` means we now have a remote shell and can run Linux commands. The message *zsh: corrupt history file* is harmless and does not affect the connection.

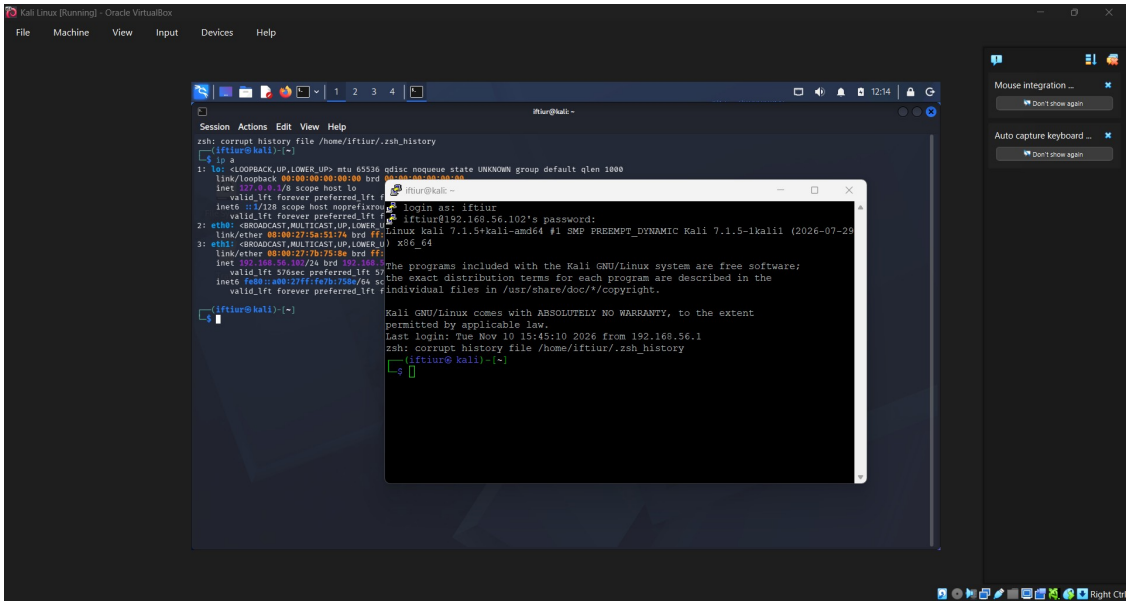


Figure 8: PuTTY logged in as `iftiur` on Kali Linux.

6.4 MobaXterm

1. Open MobaXterm Personal Edition. It starts with a local terminal; the left panel lists *User sessions* (here, WSL-Ubuntu).

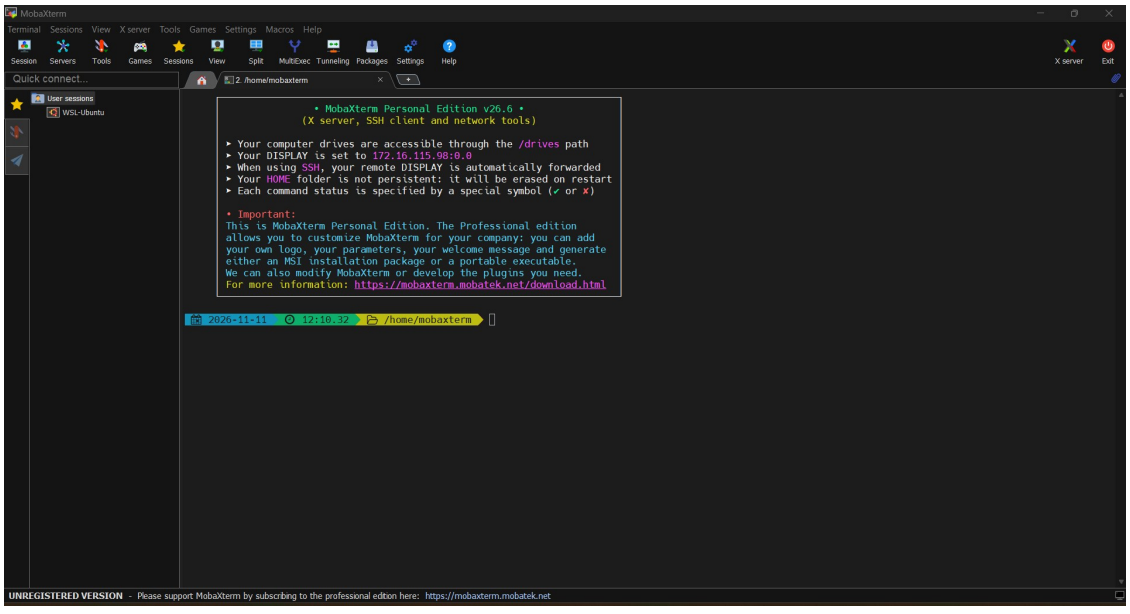


Figure 9: MobaXterm Personal Edition v26.6 start screen.

2. Click *Session* in the top toolbar, then choose *SSH*.

3. Enter the remote host `192.168.56.102`, tick *Specify username* and type `iftiur`, keep port `22`, and click *OK*.
4. Enter the password. After login, the terminal opens in a new tab and a graphical SFTP browser appears in the left panel, so files can be dragged between Windows and Kali without a separate tool.

7. Comparison of the Tools

Tool	Main purpose	Interface	Protocols used here	Best suited for
FileZilla	File transfer	Graphical, two panels	SFTP	Quick uploads and downloads
WinSCP	File transfer and light remote editing	Graphical, two panels	SFTP	Windows users who need sync and scripting
PuTTY	Remote terminal	Small window, command line	SSH	Simple, fast SSH logins
MobaXterm	All-in-one terminal, SFTP, X11	Tabbed terminal with side panel	SSH / SFTP	Administrators who manage many servers

8. Security Observations

- All four tools use SSH (port 22), so credentials and data are encrypted in transit, unlike plain FTP or Telnet.
- On first connection, each client displays the server's host key fingerprint. It should be verified before being trusted to avoid man-in-the-middle attacks (FileZilla reported *Hostkey is trusted* after acceptance).
- Key-based authentication (for example with PuTTYgen) is safer than passwords, and weak passwords should be avoided.
- The SSH server should only be enabled when it is needed, and the host-only network keeps the lab VM isolated from the outside.

9. Challenges Faced

- The VM must have the SSH service running and the correct host-only IP; otherwise the clients show *Connection refused* or *Not connected*.
- The host-only adapter must be enabled in VirtualBox so that both machines are on the same 192.168.56.0/24 network.
- The shell showed a corrupt history file warning, which can be fixed by deleting or recreating `~/.zsh_history`.

10. Conclusion

In this lab we installed FileZilla, WinSCP, PuTTY and MobaXterm and used SSH/SFTP to connect a Windows host to a Kali Linux virtual machine at 192.168.56.102. FileZilla and WinSCP proved convenient for graphical file transfer, PuTTY was the simplest tool for a remote shell, and MobaXterm offered an all-in-one environment. Understanding these tools is a basic skill for system administration, penetration testing and secure remote management.